

THARIMPEPE FEEDING SCHEME

FEEDING SCHEME MANAGEMENT SYSTEM

System Requirements Document

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Module:

XISD5319 - Work Integrated Learning
Task 2 - Requirement Analysis

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GitHub:

[ST10345327/fsms---tharimpepe: Feeding Scheme Management System](https://github.com/ST10345327/fsms---tharimpepe: Feeding Scheme Management System)

Client:

Tharimpepe Feeding Scheme
Mafikeng, North West, South Africa



Figure 1: Organization Logo

1. Requirement Analysis

1.1 Problem Domain

Tharimpepe Feeding Scheme is a community-based non-profit organisation (NPO) located in Mafikeng, North West, South Africa. The organisation provides daily meals to underprivileged children and vulnerable community members, operating through donations, volunteer support, and community partnerships.

Currently, all operational tasks are managed manually using paper-based systems and informal communication channels. This creates several critical problems for the organisation:

- Beneficiary registration is done on paper, making it difficult to track who receives meals, how often, and whether records are up to date.
- Attendance is recorded manually with no central repository, leading to inaccurate headcounts and food wastage.
- Food stock and donation tracking is done informally, making it difficult to plan meals or demonstrate impact to donors.
- Volunteer schedules are communicated via phone/WhatsApp, causing confusion, no-shows, and uneven workloads.
- Reporting to donors and community partners requires manual compilation, which is time-consuming and error-prone.

1.2 Solution Domain

The proposed solution is a web-based Feeding Scheme Management System (FSMS). Built on a three-layer MVC architecture using PHP, MySQL and Bootstrap, it will replace all manual processes with a secure, centralised digital platform providing:

1. Beneficiary Registration & Management: Digital registration with searchable records.
2. Attendance Tracking: Daily digital attendance recording per feeding session.
3. Food Stock & Donation Tracking: Real-time inventory management and donation logging.
4. Volunteer Scheduling: Online volunteer registration and schedule management.
5. Reporting & Analytics: Automated generation of reports for donors and stakeholders.
6. Secure Authentication: Role-based access control for Administrators and Volunteers.

1.3 Active Actors

The following active actors interact directly with the Feeding Scheme Management System:

Active Actor	Description	System Interaction
Organisation Manager (Admin)	Oversees all operations, approves records, generates reports	Full system access: manage beneficiaries, reports, volunteers, stock
Volunteer	Assists with meal preparation and distribution	Record attendance, view schedules, update assigned tasks
Beneficiary Representative	Guardian registering community member for meals	Submit registration form, view status
Donor	Individuals or organisations contributing food or funds	View donation records, submit donation details

1.4 Functions (Use Cases)

The following table lists all use cases of the Feeding Scheme Management System (Satzinger, Jackson and Burd, 2014):

#	Function / Use Case	Active Actor
1	Login to the system	Admin, Volunteer
2	Logout of the system	Admin, Volunteer
3	Register a new beneficiary	Admin
4	View / search beneficiary list	Admin
5	Update beneficiary details	Admin
6	Deactivate a beneficiary	Admin
7	Record daily attendance	Admin, Volunteer
8	View attendance report	Admin
9	Add food stock / donation	Admin
10	View current stock levels	Admin, Volunteer
11	Update stock levels	Admin
12	Record a donation	Admin
13	View donation history	Admin
14	Register a volunteer	Admin
15	View volunteer schedule	Admin, Volunteer

16	Assign volunteer to session	Admin
17	Generate attendance report	Admin
18	Generate stock report	Admin
19	Generate donation report	Admin
20	Generate impact report	Admin

1.5 Passive Actors

Passive actors are the systems/subsystems acted upon by active actors:

Passive Actor / System	Description
Beneficiary Management Subsystem	Stores and manages all beneficiary registration and profile data
Attendance Tracking Subsystem	Records and retrieves daily attendance data per beneficiary and session
Food Inventory Subsystem	Manages stock levels, food item records, and low-stock alerts
Donation Management Subsystem	Logs donations (food and financial), links to donor records
Volunteer Management Subsystem	Stores volunteer profiles, assigned sessions, and availability data
Reporting Subsystem	Generates reports from aggregated system data for admin use
Authentication Subsystem	Manages user login, session tokens, and role-based access control
Database (MySQL)	Central persistent data store for all system entities and records

1.6 Use Case Diagram

The following use case diagram illustrates all interactions between active actors (Admin, Volunteer, Donor) and the system use cases, enclosed within the system boundary box:



Figure 1: Use Case Diagram – Feeding Scheme Management System

2. Logical System Model

The logical system model maps each system function to its GUI inputs, outputs, system processes, and the corresponding entity relationship (database) table.

GUI Input	GUI Output	System Process	ER Table
Login credentials	Dashboard / Error message	Authenticate user	Users
Beneficiary details (name, DOB, address, guardian)	Beneficiary record confirmation	Register new beneficiary	Beneficiaries
Search keyword / ID	Beneficiary profile	Search/view beneficiary	Beneficiaries
Updated beneficiary fields	Updated record confirmation	Update beneficiary details	Beneficiaries
Session date, beneficiary list	Attendance confirmation	Record daily attendance	Attendance
No input	Attendance report (by date/period)	View/generate attendance report	Attendance
Food item name, quantity, expiry, supplier	Stock record confirmation	Add stock / donation item	FoodStock, Donations
No input	Current stock levels list	View food stock	FoodStock
Updated stock quantities	Updated stock confirmation	Update stock levels	FoodStock
Donation details (donor, amount/item, date)	Donation record confirmation	Record donation	Donations
No input	Donation history list	View donation records	Donations
Volunteer details (name, contact, availability)	Volunteer record confirmation	Register volunteer	Volunteers
Session date, volunteer ID	Schedule confirmation	Assign volunteer to session	VolunteerSessions
No input	Volunteer schedule list	View volunteer schedules	VolunteerSessions
Report type, date range	Generated report (PDF/screen)	Generate report	All tables

2.1 Input Specifications

Input Field	Data Type	Validation Rules	Required
Beneficiary First Name	String (max 50)	Letters only, no numbers	Yes
Beneficiary Last Name	String (max 50)	Letters only, no numbers	Yes
Date of Birth	Date (DD/MM/YYYY)	Must be a valid past date	Yes
Gender	Enum (Male/Female/Other)	Must select one option	Yes
Guardian Name	String (max 100)	Letters only	No
Contact Number	String (10 digits)	Numeric, 10 digits, SA format	Yes
Physical Address	String (max 200)	Free text	Yes
Attendance Date	Date	Cannot be future date	Yes
Beneficiary Present (Y/N)	Boolean	Must be True or False	Yes
Food Item Name	String (max 100)	Free text	Yes
Quantity	Decimal	Must be greater than 0	Yes
Unit of Measure	Enum (kg/litres/units)	Must select one	Yes
Donation Type	Enum (Food/Cash/Other)	Must select one	Yes
Donor Name / Organisation	String (max 100)	Free text	Yes
Volunteer Name	String (max 100)	Letters only	Yes
Username	String (max 50)	Alphanumeric	Yes
Password	String (min 8 chars)	At least 1 uppercase, 1 number	Yes

2.2 Output Specifications

Output	Format	Audience	Frequency
Beneficiary Registration Confirmation	On-screen message	Admin	On demand
Beneficiary List	Paginated table (screen)	Admin	On demand
Daily Attendance Record	On-screen confirmation + stored record	Admin, Volunteer	Daily
Attendance Report (by date/period)	Table / downloadable PDF	Admin, Donors	Weekly/Monthly
Current Stock Levels	Dashboard table	Admin, Volunteer	On demand
Low Stock Alert	On-screen notification	Admin	Automatic
Donation History Report	Table / downloadable PDF	Admin, Donors	On demand
Volunteer Schedule	Calendar view / table	Admin, Volunteer	On demand
Impact Report (meals served, beneficiaries)	Summary dashboard + PDF	Admin, Donors	Monthly
Login / Access Error Message	On-screen alert	All users	On event

2.3 System Processes

Process ID	Process Name	Description
P01	User Authentication	Validates credentials against Users table; creates a session token; enforces role-based access.
P02	Beneficiary Registration	Validates and stores new beneficiary data in the Beneficiaries table; generates a unique Beneficiary ID.
P03	Beneficiary Search & Retrieval	Queries the Beneficiaries table by name, ID, or address and returns matching records.
P04	Attendance Recording	Logs presence/absence of each beneficiary per feeding session; links to session date and Volunteer ID.

P05	Attendance Report Generation	Aggregates Attendance table records by date range; calculates total meals served and average attendance.
P06	Food Stock Management	Updates FoodStock table on addition or consumption; triggers low-stock alert when quantity falls below threshold.
P07	Donation Recording	Stores donation details in the Donations table; links donation to a donor and optionally to a stock item.
P08	Volunteer Registration & Scheduling	Creates volunteer record; assigns volunteer to sessions via the VolunteerSessions linking table.
P09	Report Generation	Aggregates data across tables to produce formatted reports for admin or donor audience.
P10	Data Validation	Cross-cutting process: validates all input fields before database write operations.

3. Entity Relationship Tables

The following tables define the entities and their attributes in the Feeding Scheme Management System database.

3.1 Users Table

Field Name	Data Type	Constraints	Description
UserID	INT	PK, AUTO INCREMENT, NOT NULL	Unique identifier for each user
Username	VARCHAR(50)	UNIQUE, NOT NULL	Login username
PasswordHash	VARCHAR(255)	NOT NULL	Bcrypt-hashed password
FullName	VARCHAR(100)	NOT NULL	Full name of the user
Email	VARCHAR(100)	UNIQUE, NOT NULL	User email address
Role	ENUM('Admin','Volunteer')	NOT NULL, DEFAULT 'Volunteer'	Role determines system access level
IsActive	BOOLEAN	NOT NULL, DEFAULT TRUE	Whether the account is active
CreatedAt	DATETIME	NOT NULL, DEFAULT NOW()	Account creation timestamp

3.2 Beneficiaries Table

Field Name	Data Type	Constraints	Description
BeneficiaryID	INT	PK, AUTO INCREMENT, NOT NULL	Unique beneficiary identifier
FirstName	VARCHAR(50)	NOT NULL	Beneficiary first name
LastName	VARCHAR(50)	NOT NULL	Beneficiary last name
DateOfBirth	DATE	NOT NULL	Date of birth

Gender	ENUM('Male','Female','Other')	NOT NULL	Gender of beneficiary
GuardianName	VARCHAR(100)	NULL	Name of guardian (if minor)
ContactNumber	VARCHAR(15)	NOT NULL	Contact phone number
Address	VARCHAR(200)	NOT NULL	Physical address
RegistrationDate	DATE	NOT NULL, DEFAULT CURDATE()	Date of registration
IsActive	BOOLEAN	NOT NULL, DEFAULT TRUE	Whether beneficiary is currently active
RegisteredBy	INT	FK -> Users(UserID)	Admin who registered this beneficiary

3.3 Attendance Table

Field Name	Data Type	Constraints	Description
AttendanceID	INT	PK, AUTO INCREMENT, NOT NULL	Unique attendance record ID
BeneficiaryID	INT	FK -> Beneficiaries(BeneficiaryID), NOT NULL	Beneficiary who attended
SessionDate	DATE	NOT NULL	Date of the feeding session
IsPresent	BOOLEAN	NOT NULL	TRUE = present, FALSE = absent
RecordedBy	INT	FK -> Users(UserID)	User who recorded attendance
Notes	TEXT	NULL	Optional notes (e.g. illness, late arrival)

3.4 FoodStock Table

Field Name	Data Type	Constraints	Description
StockID	INT	PK, AUTO INCREMENT, NOT NULL	Unique stock item ID
ItemName	VARCHAR(100)	NOT NULL	Name of the food item
Category	VARCHAR(50)	NOT NULL	Category (Grain, Protein, Vegetable, etc.)
Quantity	DECIMAL(10,2)	NOT NULL, DEFAULT 0	Current quantity available
UnitOfMeasure	ENUM('kg','litres','units')	NOT NULL	Unit of measure for quantity
MinimumThreshold	DECIMAL(10,2)	NOT NULL, DEFAULT 5	Minimum quantity before alert triggered
ExpiryDate	DATE	NULL	Expiry date of batch
UpdatedBy	INT	FK -> Users(UserID)	User who last updated this record

3.5 Donations Table

Field Name	Data Type	Constraints	Description
DonationID	INT	PK, AUTO INCREMENT, NOT NULL	Unique donation ID
DonorName	VARCHAR(100)	NOT NULL	Name of donor or organisation
DonationType	ENUM('Food','Cash','Other')	NOT NULL	Type of donation
ItemDescription	VARCHAR(200)	NULL	Description of food

			donated (if food)
Quantity	DECIMAL(10,2)	NULL	Quantity donated (if food)
AmountRand	DECIMAL(10,2)	NULL	Amount in Rands (if cash)
DonationDate	DATE	NOT NULL, DEFAULT CURDATE()	Date donation was received
LinkedStockID	INT	FK -> FoodStock(StockID), NULL	Food stock item updated by this donation
RecordedBy	INT	FK -> Users(UserID)	Admin who recorded this donation

3.6 Volunteers Table

Field Name	Data Type	Constraints	Description
VolunteerID	INT	PK, AUTO INCREMENT, NOT NULL	Unique volunteer ID
UserID	INT	FK -> Users(UserID), NULL	Linked system user account (if applicable)
FullName	VARCHAR(100)	NOT NULL	Full name of volunteer
ContactNumber	VARCHAR(15)	NOT NULL	Contact phone number
Email	VARCHAR(100)	NULL	Email address
Availability	TEXT	NULL	Notes on availability
IsActive	BOOLEAN	NOT NULL, DEFAULT TRUE	Whether volunteer is currently active

RegisteredDate	DATE	NOT NULL, DEFAULT CURDATE()	Date volunteer was registered
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3.7 VolunteerSessions Table

Field Name	Data Type	Constraints	Description
SessionID	INT	PK, AUTO INCREMENT, NOT NULL	Unique session record ID
VolunteerID	INT	FK -> Volunteers(VolunteerID), NOT NULL	Volunteer assigned to session
SessionDate	DATE	NOT NULL	Date of the feeding session
Role	VARCHAR(100)	NULL	Volunteer role for this session (e.g. Food Prep)
AttendedSession	BOOLEAN	NULL, DEFAULT NULL	Whether volunteer attended (confirmed after session)

3.8 Entity Relationship Diagram (ERD)

The following ERD illustrates all seven entities and their relationships (Primary Keys, Foreign Keys, and cardinality) for the Feeding Scheme Management System database:

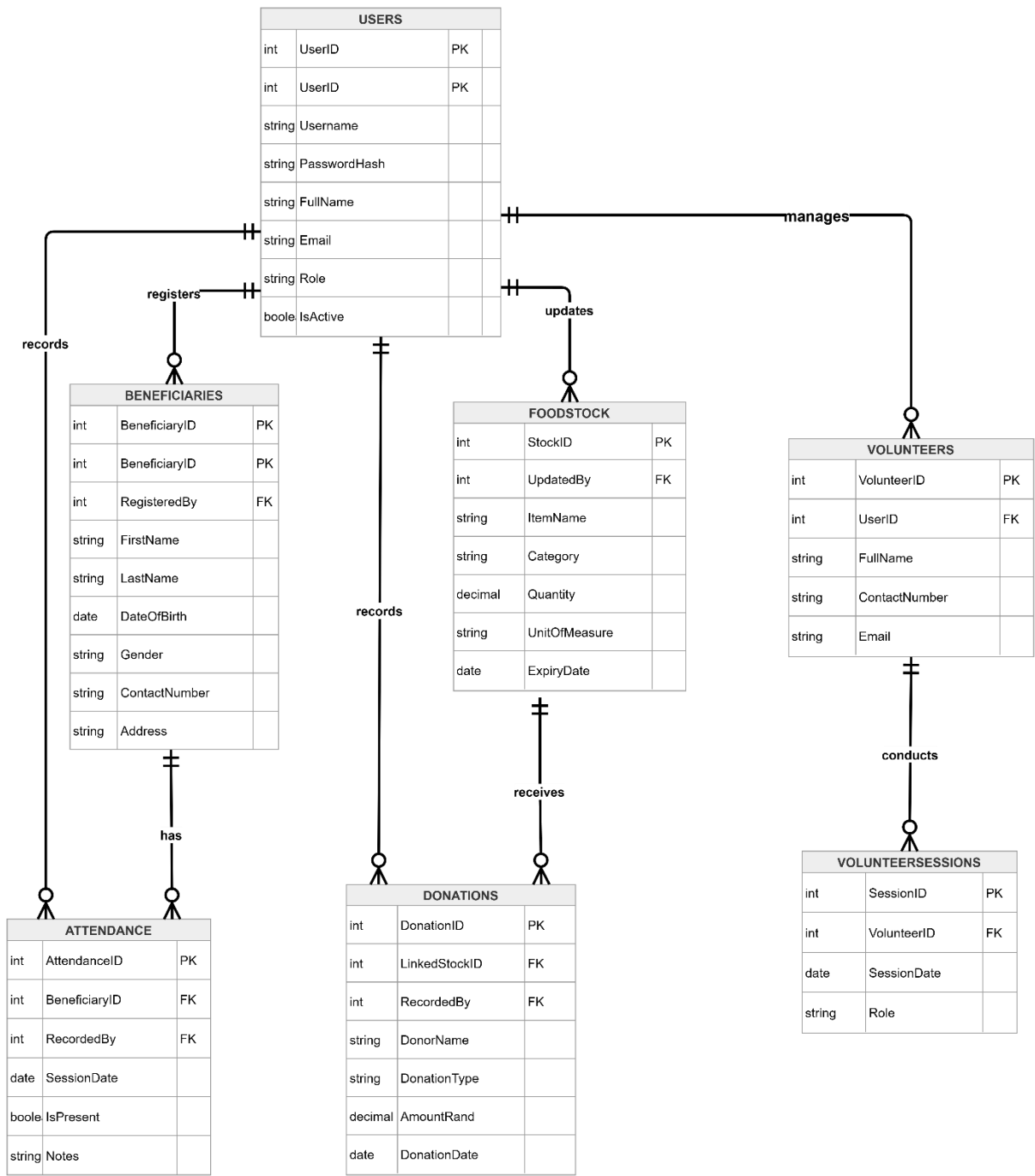


Figure 2: Entity Relationship Diagram (ERD) – Feeding Scheme Management System

3.9 Business Rules & Entity Relationships

Entity	Related Entity	Relationship	Business Rule
User (Admin)	Beneficiary	One-to-Many	One admin can register many beneficiaries. Each beneficiary is registered by one admin.
Beneficiary	Attendance	One-to-Many	One beneficiary can have many attendance records. Each record belongs to one beneficiary.
User	Attendance	One-to-Many	One user can record many attendance entries. Each entry is recorded by one user.
FoodStock	Donation	One-to-Many (opt)	One stock item can be linked to many donations. A donation may or may not update a stock item.
Volunteer	VolunteerSession	One-to-Many	One volunteer can be assigned to many sessions.
User	Volunteer	One-to-One (opt)	A volunteer may optionally have a system user account.

4. Class Diagram

The following UML Class Diagram shows all system classes with their attributes, methods, and relationships. Inheritance arrows indicate that Admin and Volunteer are subclasses of the User base class.

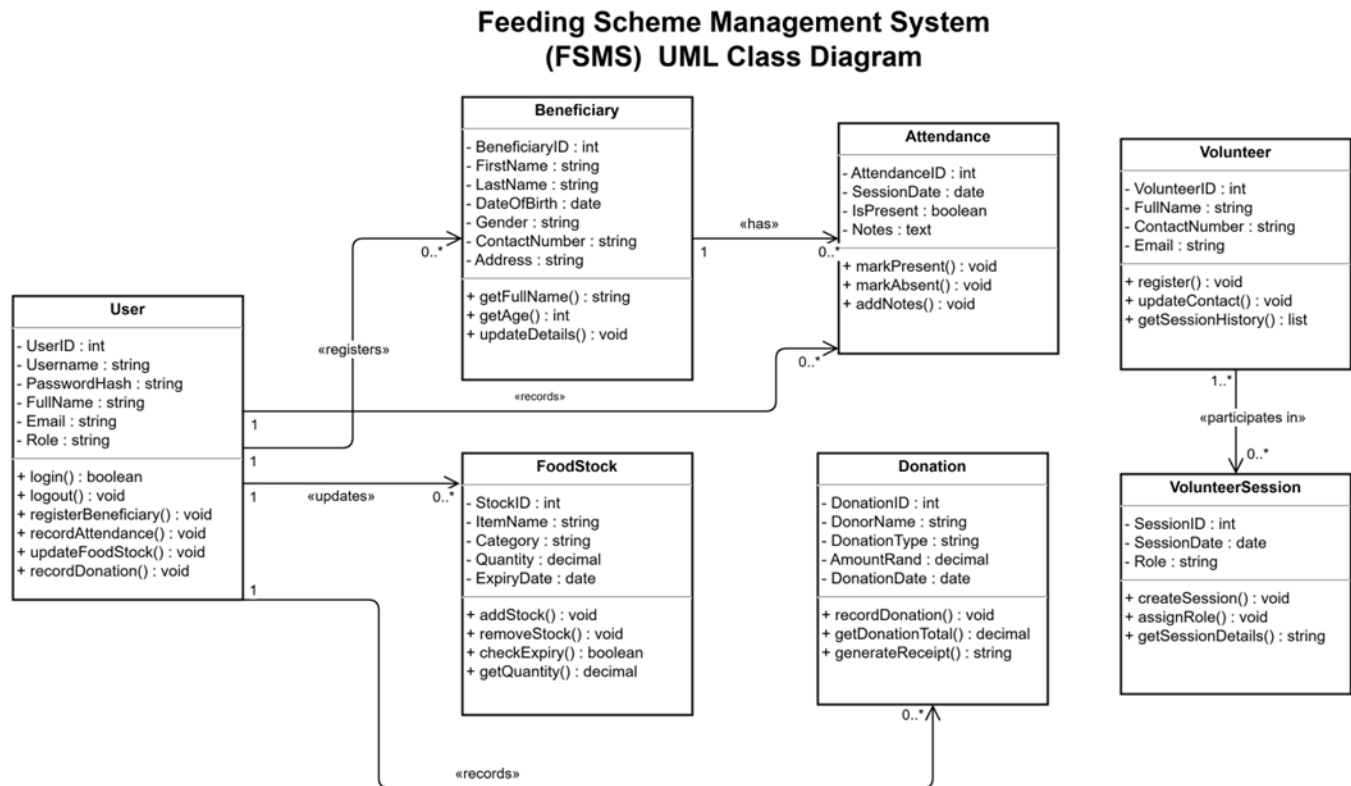


Figure 3: UML Class Diagram – Feeding Scheme Management System

4.1 Class Descriptions

Class	Key Attributes	Key Methods	Relationships
User	userID, username, passwordHash, role	login(), logout(), updateProfile()	Parent of Admin and Volunteer
Admin (extends User)	adminLevel	registerBeneficiary(), generateReport()	Manages Beneficiary, FoodStock, Donation
Volunteer (extends User)	availability, sessions	recordAttendance(), viewSchedule()	Linked to VolunteerSession
Beneficiary	beneficiaryID, firstName, dateOfBirth, address	register(), update(), deactivate()	Has many Attendance records

Attendance	attendanceID, sessionDate, isPresent	record(), getByDate(), getByBeneficiary()	Belongs to Beneficiary
FoodStock	stockID, quantity, minThreshold	addStock(), deductStock(), checkLowStock()	Linked to Donation
Donation	donationID, donorName, amountRand	record(), linkToStock()	May link to FoodStock
VolunteerProfile	volunteerID, fullName, availability	register(), assignToSession()	Has many VolunteerSessions
Report	reportType, dateRange	generate(), export()	Aggregates data from all classes

5. Functional & Non-Functional Requirements

5.1 Functional Requirements

- FR01: The system must allow admin users to register new beneficiaries with all required personal details.
- FR02: The system must allow admin and volunteer users to record daily attendance per feeding session.
- FR03: The system must track food stock levels and alert the admin when any item falls below the minimum threshold.
- FR04: The system must allow admin users to record food and cash donations and link them to stock items where applicable.
- FR05: The system must allow admin users to register volunteers and assign them to feeding sessions.
- FR06: The system must generate attendance, stock, donation, and impact reports for admin use.
- FR07: The system must enforce role-based access control – Admins have full access; Volunteers have limited access.
- FR08: The system must allow admin users to search for beneficiaries, volunteers, and donations using keywords.
- FR09: The system must allow admin users to update or deactivate beneficiary and volunteer records.
- FR10: The system must display a dashboard summary (total beneficiaries, meals served, stock levels) on login.

5.2 Non-Functional Requirements

- NFR01 – Usability: The interface must be simple and intuitive for users with limited digital literacy.
- NFR02 – Availability: The system must be available 24/7 with maximum 1% planned downtime.
- NFR03 – Security: All passwords must be hashed. All data transmission must use HTTPS/SSL.
- NFR04 – Performance: Page load times must not exceed 3 seconds under normal conditions.
- NFR05 – Reliability: The system must support offline data entry for attendance when internet is unavailable.
- NFR06 – Maintainability: The codebase must follow MVC architecture to facilitate future enhancements.
- NFR07 – Scalability: The system must support up to 500 beneficiaries and 50 users without degradation.
- NFR08 – Compatibility: The system must function correctly on Chrome, Firefox, and Edge browsers.

6. References

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